

Norma 0

$$\|a\|_0 \neq \left(\sum_{i=1}^n |a_i|^0 \right)^{1/0}$$

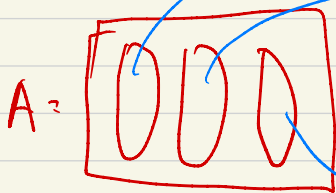
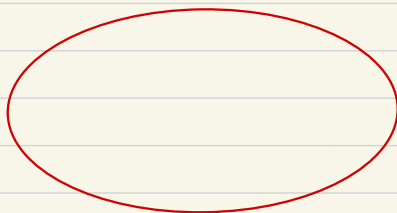
$$\|a\|_0 = \sum_{i=1}^n \mathbb{I}(a_i \neq 0)$$

$$\|c \cdot a\|_0 \neq |c| \cdot \|a\|_0 \Rightarrow \| \cdot \|_0 \text{ is not a norm}$$

$$\begin{bmatrix} 0.5 \\ 0 \\ 1/3 \\ 4 \end{bmatrix}$$

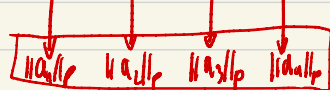
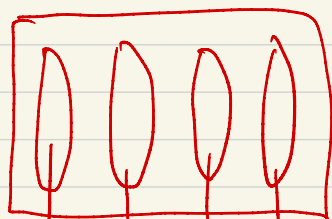


ponderado



$$\text{vec}(A) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$A :$



$$= b^T \Rightarrow$$

$$\|b\|_q = \|A\|_{p,q}$$

$$Ax = \lambda x$$

$$\lambda x - \overset{\text{Id}}{Ax} = 0 \Rightarrow (\lambda I - A)x = 0$$

$$\bullet (\lambda I - A)^{-1} \Rightarrow x = (\lambda I - A)^{-1} 0 = 0$$

$$\bullet \det(\lambda I - A) = 0$$

Problem 1

$$\text{tr}(A) = \sum_{i=1}^n \lambda_i$$

$$\text{tr}(A) = \text{tr}(X \overset{I}{\Delta} X^{-1}) = \text{tr}(\overset{I}{X^{-1}X} \cdot \Delta) = \text{tr}(\Delta) = \sum_{i=1}^n \lambda_i$$

$$\det(A) = \prod_{i=1}^n \lambda_i$$

$$\det(A) = \det(X \Delta X^{-1}) = \det(X) \cdot \det(\Delta) \underbrace{\det(X^{-1})}_{1/\det(X)} = \det(\Delta)$$

$$= \prod_{i=1}^n \lambda_i$$

Problem 2

$$\|A\|_F = \sqrt{\sum_i \sigma_i^2}$$

$$\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\text{tr}(A^T A)} \stackrel{A=U\Sigma V^T}{=} \sqrt{\text{tr}[(U\Sigma V^T)^T (U\Sigma V^T)]}$$

$$= \sqrt{\text{tr}(V \Sigma^T \underbrace{V^T U}_{I} \Sigma V^T)} = \sqrt{\text{tr}(V \Sigma^T \Sigma V^T)} = \sqrt{\text{tr}(\underbrace{V^T V}_{I} \Sigma^T \Sigma)}$$

$$= \sqrt{\text{tr}(\Sigma^T \Sigma)} = \sqrt{\sum_i \sigma_i^2}$$

Problem 3

$$A = U \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} V^T$$

$$A^T = V \begin{bmatrix} \Sigma & 0 \end{bmatrix} U^T$$

$$A^\# = V \begin{bmatrix} \Sigma^{-1} & 0 \end{bmatrix} U^T$$

$$AA^\# = U \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} \underbrace{V^T V}_I \begin{bmatrix} \Sigma^{-1} & 0 \end{bmatrix} U^T$$

$$= U \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} \begin{bmatrix} \Sigma^{-1} & 0 \end{bmatrix} U^T = U \begin{bmatrix} \Sigma \Sigma^{-1} & \Sigma \cdot 0 \\ 0 \cdot \Sigma & 0 \cdot 0 \end{bmatrix} U^T$$

$$= U \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} U^T$$

$$X = \text{diag}(1, \dots, 1, 0, \dots, 0)$$

$$\begin{aligned}
 AA^T &= V \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} \underbrace{V^T V}_I \begin{bmatrix} \Sigma & 0 \end{bmatrix} V^T \\
 &= V \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} \begin{bmatrix} \Sigma & 0 \end{bmatrix} V^T \\
 &= V \begin{bmatrix} \Sigma^2 & 0 \\ 0 & 0 \end{bmatrix} V^T \\
 &\quad \underbrace{\hspace{10em}}_{X = \text{diag}(\sigma_1^2, \dots, \sigma_m^2, 0, \dots, 0)}
 \end{aligned}$$

$$\begin{aligned}
 I - AA^T &= I - V \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix} V^T = V X V^T \\
 &= V I V^T - V \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix} V^T \\
 &= V \left(I - \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix} \right) V^T \\
 &= V \left(\begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} - \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix} \right) V^T \\
 &= V \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} V^T \\
 &\quad \underbrace{\hspace{10em}}_{X = \text{diag}(0, \dots, 0, 1, \dots, 1)}
 \end{aligned}$$